

What we're doing

How good was the best contacts-v1 model we had on any given date — on contact R-precision and on held-out validation loss — and how tightly do those two track each other?

Every number exists somewhere already (#67, #75, #82, #89, #108, #117, #120, #146, #155, #160, #169), but they are scattered across issue comments, per-experiment CSVs and two W&B projects, with three different inference recipes mixed in. This experiment collects them into one dataset and three standing figures, and keeps them refreshable as new models land.

Why

Not a hypothesis-testing experiment — it is a **standing tracker**. The two things it is built to make visible, both of which prior issues assert piecemeal:

1. Progress on contact accuracy has come almost entirely from the **base model**, not from inference or post-training. 2. Validation loss predicts contact accuracy **across training generations** and stops predicting it **within a generation** (#169's finding), so the loss frontier and the accuracy frontier are not the same curve.

Results so far

The accuracy frontier moved in three jumps, all from the base model, none from inference or post-training: ~0.03 to 0.425 when #75's E8 rung finished (2026-06-21), 0.436 to 0.534 when #117's 16-epoch bs256 run finished (2026-07-22), and 0.534 to 0.556 across #155's crops+contacts-v1+ESM-Atlas 3-way mixture restart, which finished training 2026-08-01 (peak at its step 70000 checkpoint; the final checkpoint reads 0.554, a 0.002 dip inside noise). #155 is the first jump from a *data* change rather than a hyperparameter sweep or epoch count.

Between the first two, five weeks of post-training and inference work moved it by +0.011 (#120's re-epoch), and #160's backtracking fine-tune moved it by -0.020.

A new diagnostic on #155's final checkpoint: scoring each of its 100 sampled rollouts per protein on its own best contacts, instead of voting them together, reads 0.595 — +0.041 of headroom that isn't reachable without ground truth, but points at reranking/selection as an unexplored lever.

Structure predictors on the same 554 proteins: Protenix-v2 single-seq 0.603, ESMFold 0.755, ESMFold2 0.786, Protenix-v2 + MSA 0.812. We are still below all of them (the oracle ceiling alone would essentially close the gap to Protenix-v2 single-seq, but it isn't a deliverable number).

Loss tracks accuracy across generations, not inside one

The 0.053-nat #75 to #117 gap buys +0.109 R-precision — about 2 R-precision per nat. The 0.008-nat gap between #117's early-stop and final checkpoints buys nothing, and #146's 3B is 0.0012 better on loss and 0.023 worse on R-precision.

So the loss frontier is useful for deciding which checkpoints are worth scoring, and useless for picking between two checkpoints of one run.

The trap in these numbers

The same weights score ~ 0.086 higher under exp82's rollout recipe than under exp89's original pairwise scorer (#61/#75 E8: 0.339 to 0.425; #120: 0.350 to 0.436). That is comparable to two generations of model progress.

The figures keep pairwise and rollout visually separate. Anything from the eval-checkpoint skill is pairwise; anything from exp82's rollout workers or the exp169 dispatcher is rollout. Never infer the recipe from the magnitude.

A third recipe, oracle best-of-100, is a diagnostic upper bound, not a deployable score (see Results so far) — it never enters the "best model trained to date" frontier line or headline labelling, only its own marker.

Open gap

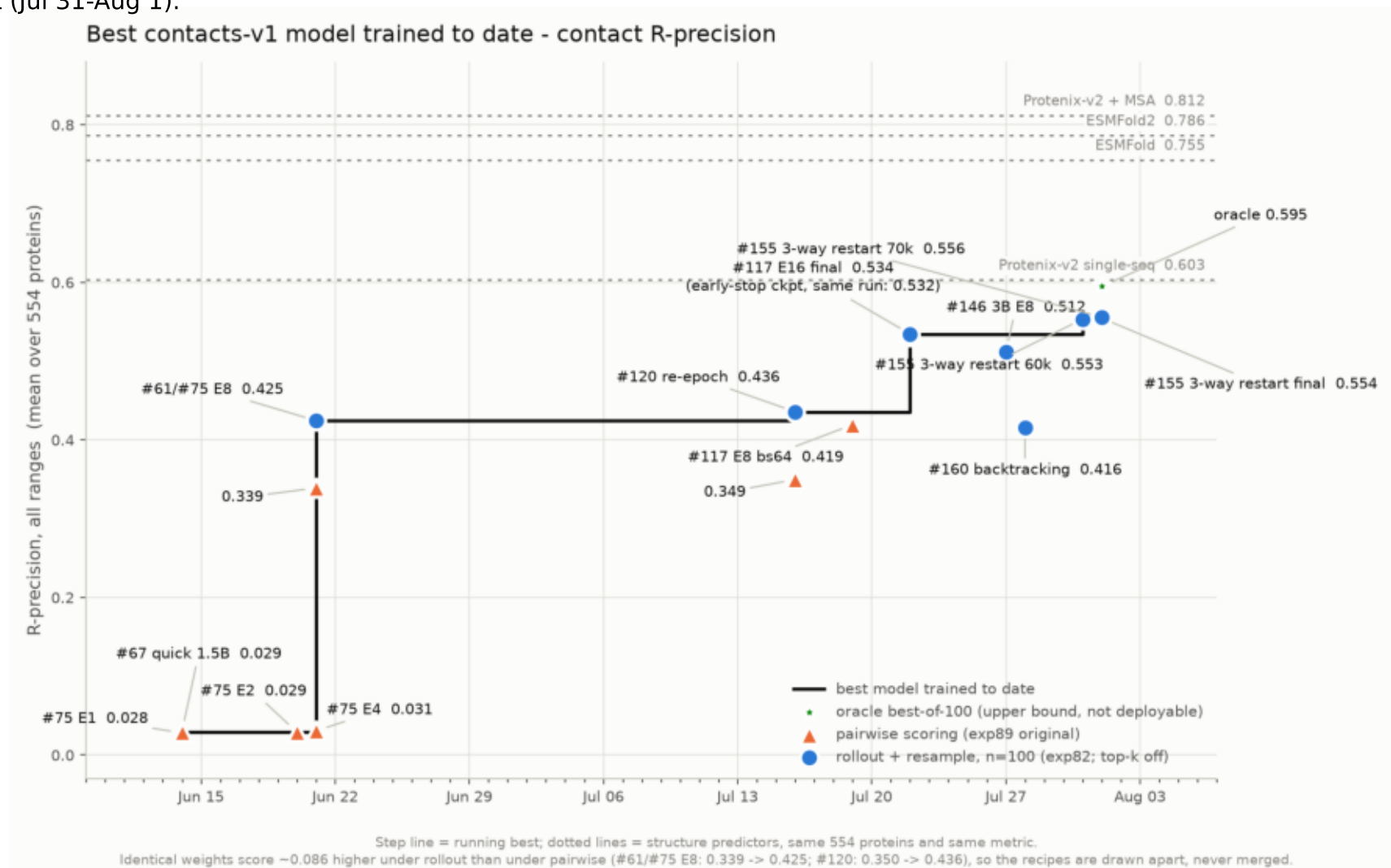
#108's 3B on CoreWeave held the loss frontier from 2026-07-11 to 07-16 at 2.7418 and was never contact-scored. Given #146 — a 3B at matched loss under-performing the 1.5B — it is probably not a missed frontier point, but that is an inference, not a measurement.

Keeping this current

Two commands: `build_dataset.py` re-pulls W&B, `plot_progress.py` redraws. New benchmark scores are hand-added to `RPRECISION_ROWS` with a source citation and an explicit inference recipe. Full procedure in the README.

rprecision_frontier.png

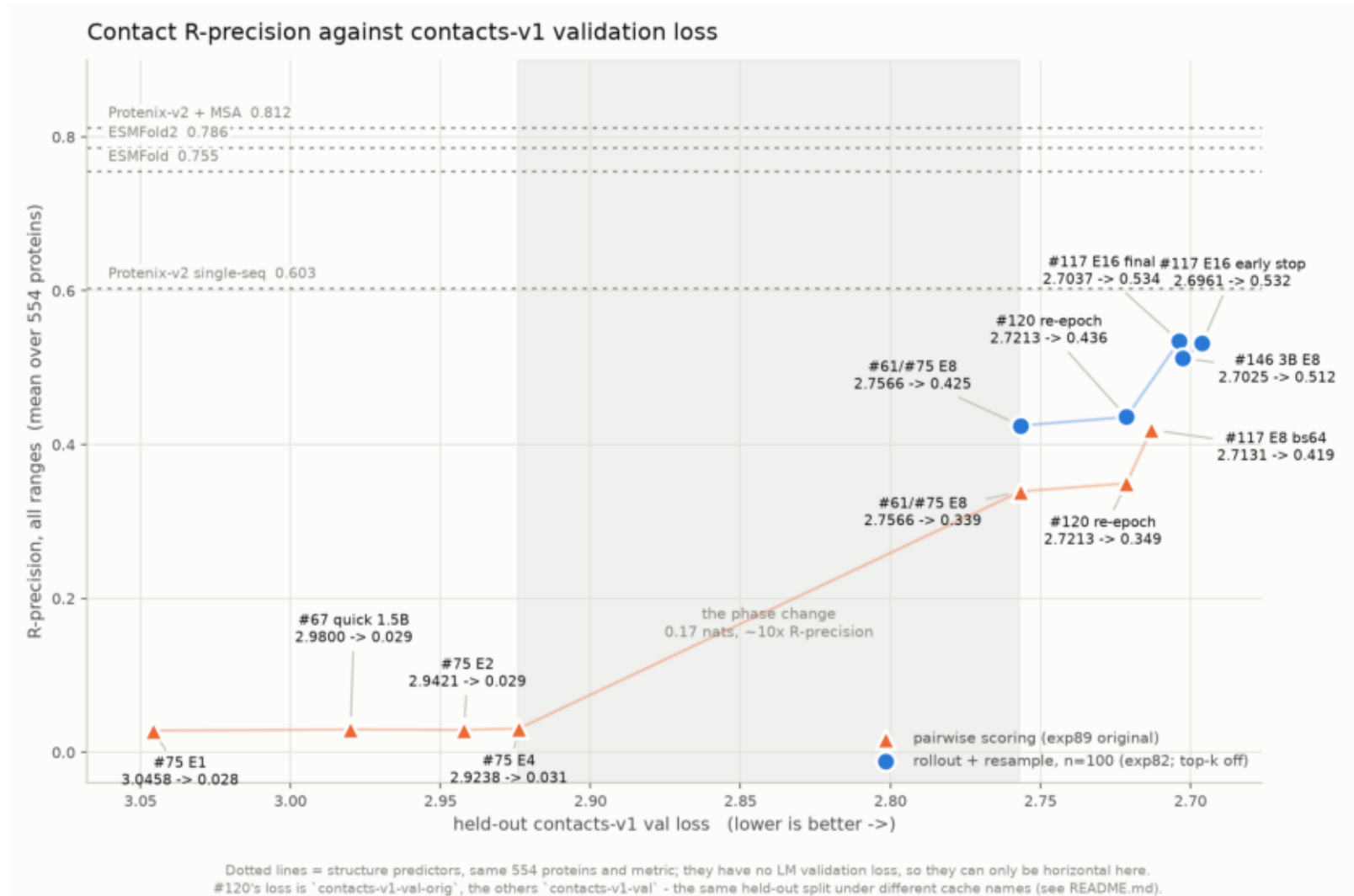
Best contacts-v1 model to date on contact R-precision. Three jumps, all from the base model: #75 E8 (Jun 21), #117 E16 (Jul 22), #155 3-way restart (Jul 31-Aug 1).



\$ plot_progress.py

rprecision_vs_val_loss.png

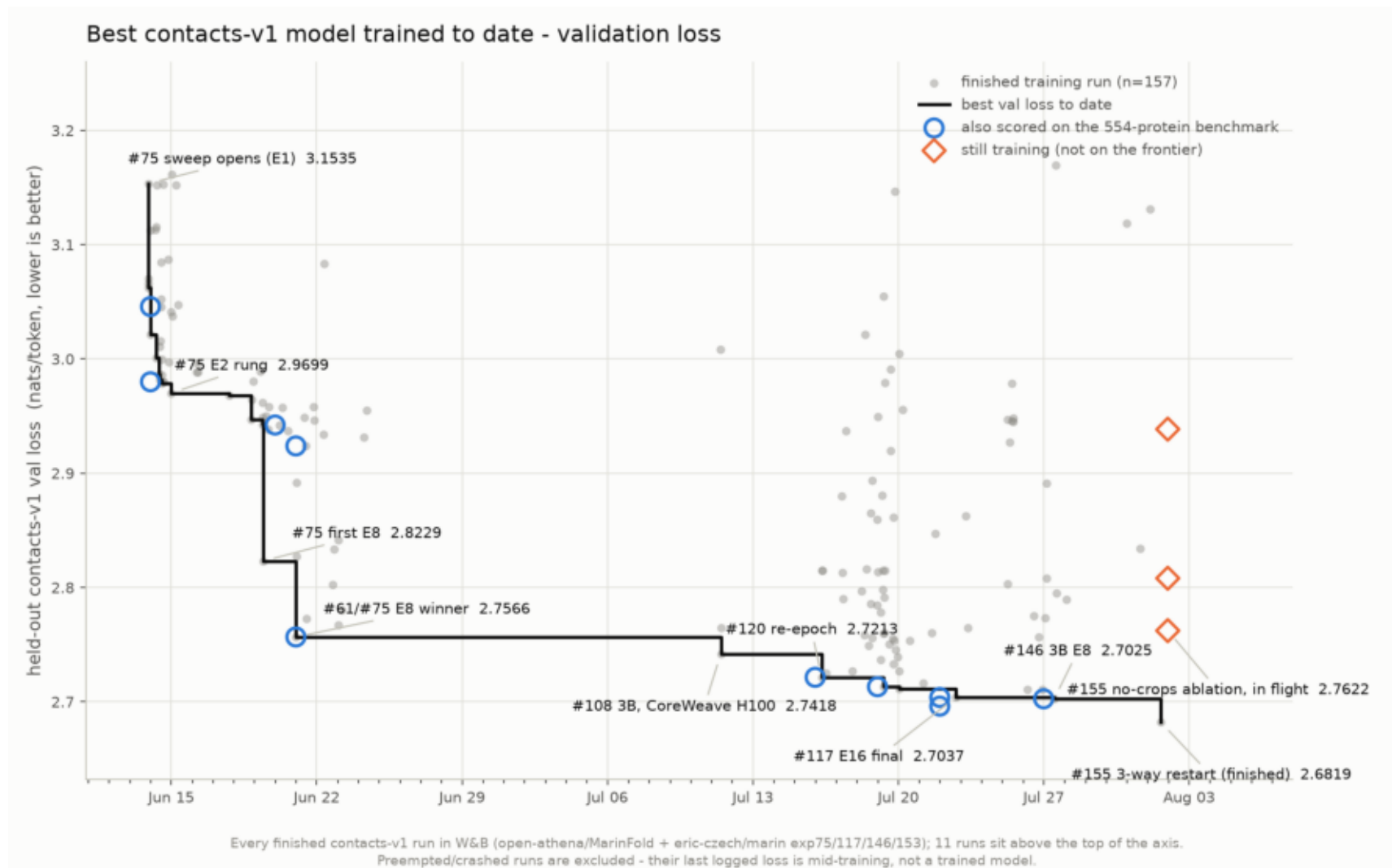
R-precision against val loss. ~2 R-precision per nat across generations; flat inside one run (#117 early vs final).



\$ plot_progress.py

val_loss_frontier.png

Best held-out contacts-v1 val loss to date over 157 finished runs. The loss frontier has many more steps than the accuracy one.



\$ plot_progress.py